

VR Trainee: a Virtual Reality Training Tool for the Energy Industry

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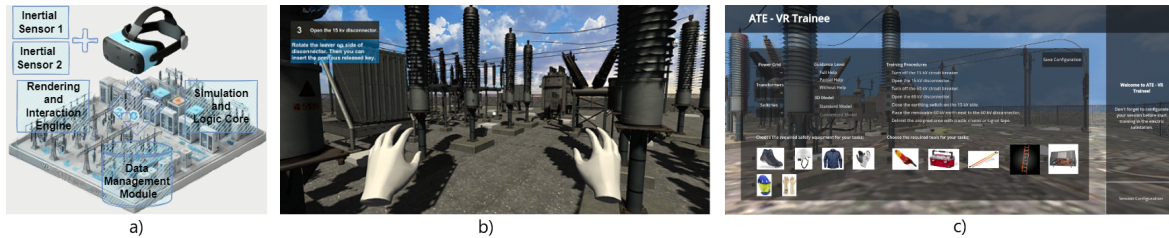


Figure 1: a) VR Trainee's architecture; b) User's hands in virtualized environment in VR Trainee; c) VR Trainee's customization.

ABSTRACT

Operating hazardous machinery often requires comprehensive training, which can be costly, time-consuming, and lead to delays. To address these challenges, we introduce VR Trainee, a Virtual Reality-based tool designed for electrical substation workers. VR Trainee allows users to interact with objects in a safe, controlled environment, enhancing training effectiveness for new maintenance employees. Equipped with inertial sensors, it gathers data to provide real-time feedback related to users' actions. Initial validation shows significant skill improvements, thanks to trial-and-error opportunities and clear instructions. Our work demonstrates VR's potential to streamline learning and boost user motivation.

Index Terms: Substation, High-risk Tasks, Worker Training, Virtual Reality, Immersive Application, Tracking and Sensing.

1 INTRODUCTION

High-voltage substations present significant safety risks during maintenance activities, requiring specialized training to minimize errors and ensure operational safety. Traditional training methods, however, often need more immersive, hands-on experience required for effective skill development, leaving trainees needing more preparation for real-world scenarios. VR Trainee addresses these challenges by providing an innovative virtual reality training tool that simulates realistic substation maintenance tasks. This immersive platform allows trainees to engage safely in practical exercises, refining their skills and building confidence in a controlled environment. The primary goals of VR Trainee are to improve safety and competence in substation maintenance through virtual training, enabling users to learn from their mistakes without the inherent risks of live operations. Due to page limitation, we only present three solutions that resort to VR for training purposes [1] [2] [4]. However, when comparing VR Trainee to a detailed view of the state of the art, it stands out by not only enhancing training through the use of inertial sensors but also by employing a modular design that allows it to be extended to simulate other specialized scenarios (training environments, equipment, procedures, etc.) in different

domain other than the energy sector. VR Trainee is being developed as part of the Alliance for the Energy Transition (ATE) project that targets the decarbonization, decentralization and digitalization of the Energy sector in Portugal, and aims to allow the energy industry to train workforce, and even clients, through a cost-effective and flexible alternative compared to traditional training methods.

2 SYSTEM ARCHITECTURE

VR Trainee's architecture (cf., Figure 1a) is designed to facilitate an immersive and effective training experience. It is based on a modular, layered structure that ensures scalability, cross-platform compatibility, and real-time user interaction processing. It comprises three core components: i) **Rendering and Interaction Engine**: responsible for the visualization of the virtual substation environment and managing all interactions between the user and the simulation. It handles the 3D representation of substation equipment, ensuring high fidelity and realism. The engine interfaces directly with VR hardware, supporting features such as hand-tracking and gesture recognition for a seamless user experience. Optimized for real-time feedback, it provides immediate visual and auditory cues in response to user actions; ii) **Simulation and Logic Core**: govern the internal operations of VR Trainee, executing the business logic and rules that dictate the behaviour of substation components. It simulates equipment operations, faults, and maintenance procedures, processing user inputs to trigger system events. This component ensures simulation consistency and stability, managing physics calculations and safety mechanisms critical for accurate training experiences; and iii) **Data Management Module**: responsible for the storage and retrieval of needed information. This module tracks training progress and session data. It logs user interactions for performance analysis while ensuring secure handling of sensitive information. The architecture's modularity allows for easy updates and the integration of advanced features, ensuring that VR Trainee evolves alongside technological advancements and training needs.

3 DESIGN PRINCIPLES

VR Trainee's design considers several key interaction design principles [3] to enhance usability and user experience: i) **Visibility of System Status**: the user interface provides continuous feedback, ensuring users are always aware of their current status during training. A hints interface guides users throughout the process; ii) **Match between the System and the Real World**: the interface uses simple, user-friendly language and follows real-world conventions, making it intuitive. Clear instructions are used to avoid technical jargon and facilitate understanding; iii) **Consistency and Standards**: the

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design maintains internal and external consistency. Colors and patterns are uniform across screens, while familiar icons and symbols align with user expectations, improving usability; iv) **User Control and Freedom:** users can easily correct mistakes, fostering a sense of control. However, in maintenance tasks, restrictions on error correction simulate real-world consequences to encourage learning; v) **Error Prevention:** potential errors (e.g., user interaction, technical, training and instruction, and graphics and performance) are addressed proactively to minimize occurrence, with specific dialogues reserved for critical situations; vi) **Recognition Rather than Recall:** relevant options and instructions remain visible, reducing the user's memory load. Essential information, such as substation layout, is always accessible; and vii) **Minimalist Design and Aesthetic:** a clutter-free user interface displays only essential information. This minimalist approach helps users focus on the task without being overwhelmed by unnecessary details. While these principles were effectively integrated into VR Trainee, **Flexibility and Efficiency of Use, Error Recovery Mechanisms, and Comprehensive Help Systems** are planned for future versions of VR Trainee.

4 EVALUATION

A structured evaluation process was conducted to assess the effectiveness and usability of the VR Trainee.

4.1 Usability

The usability evaluation of VR Trainee focused on three key factors: effectiveness, efficiency, and user satisfaction. The goal was to determine how well the system meets the needs of users and how intuitive and easy to use it is in a training context. Participants interacted with the system through a series of substation maintenance tasks and provided feedback on their experience through questionnaires and interviews. Metrics such as task completion time, error rates, and user satisfaction were gathered to assess the system's performance. The feedback revealed areas where users found the system to be intuitive, as well as parts of the interface that required improvement, particularly in terms of navigation and guidance.

4.2 Testing

In addition to usability evaluation, formal testing was conducted to measure the system's overall functionality and reliability. A total of ten participants, including seven novice and three experienced with this substation maintenance workers, were selected to interact with the VR training tool. The average age of participants was 42 ± 8.5 years and the group included six men and four women. These tests were conducted in controlled environments with standardized VR equipment, which included a VR headset, inertial sensors, and a designated interaction space. A think-aloud protocol was encouraged to gather real-time feedback on user actions, thoughts, and any difficulties found during the session. The instructions performed by participants are related to high-voltage substation maintenance (cf., Figure 1b): turnoff the 15kV circuit breaker and then operate the respective disconnector, while being observed by facilitators. Each session lasted approximately 1 hour (interaction with the VR system and feedback collection). Data was collected on task completion time, error frequency, and system stability during interactions. Post-session interviews provided further insight into user experiences, focusing on comfort, perceived realism, and ease of use.

5 DISCUSSION

The usability evaluation process confirmed that VR Trainee effectively meets its design goals, providing an engaging and educational training experience for substation maintenance. Participants reported increased confidence in their skills and expressed positive feedback regarding the immersive nature of the training environment. Tests show that users find the system favorable, with an overall score of 3.6 out of 5, and easy to use, scoring 3.8 out of 5. Users

also believe that they can quickly learn the system (3.3 out of 5), and only a few found it cumbersome (1.6 out of 5). They felt confident using the system (3.4 out of 5), with eight out of ten finding the procedures easy to learn and nine out of ten considering it an advantage over traditional methods. In addition, customization (cf., Figure 1c) and instruction clarity scored high, with eight users finding them easy to understand and only two expressing confusion. This suggests that users are learning effectively, corroborated by iterative UI design improvements aimed at simplicity and clarity. Another critical observation is the need for initial technical support (scoring 1.5 out of 5 for ease of initial use), indicating that a proficient user should assist during the first setup. Despite this, system integration received high marks (4.4 out of 5), and the lack of inconsistencies was positively noted (1.6 out of 5). One interesting observation is that users felt a need to learn many things before starting (3.9 out of 5). This could be due to an unfamiliarity with virtual reality technology and substation-specific terminology. However, the system's ease of learning helps mitigate this initial hurdle. Data collection, enabled by inertial sensors, proved vital to the training process as they allowed for better capturing of users' motion and helped provide real-time feedback on user actions. Still, such motion data can be further explored to enhance user feedback and integrate performance analytics into VR Trainee.

6 CONCLUSION AND FUTURE DIRECTIONS

The development of VR Trainee marks a significant advancement in substation maintenance training. By utilizing immersive virtual reality, the system addresses key challenges in traditional methods, such as accessibility, engagement, and safety. Usability testing shows that VR Trainee offers an intuitive interface and a realistic environment, improving learning outcomes and boosting user confidence. Future enhancements for VR Trainee include: i) **Expanded Training Modules:** adding more substation maintenance scenarios to cater to diverse training needs; ii) **Integration of Performance Analytics:** incorporating metrics to help trainers track progress and pinpoint areas for improvement; iii) **Mobile and Remote Access:** enabling mobile and remote access to broaden the system's reach; iv) **Gamification Elements:** adding features like leader boards and achievements to increase user engagement; and v) **Longitudinal Studies:** conducting studies to assess long-term skill retention and the system's sustained effectiveness. These improvements will help VR Trainee evolve into a comprehensive training solution that meets current needs while adapting to emerging challenges.

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REFERENCES

- [1] T. Hoang, S. Greuter, and S. Taylor. An evaluation of virtual reality maintenance training for industrial hydraulic machines. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 573–581, 2022. doi: 10.1109/VR51125.2022.00077 1
- [2] Z. Jiang, Y. Yang, Q. Yuan, P. Leng, Y. Liu, and Z. Pan. Virtual reality training environment for electric systems. In *2021 IEEE 7th International Conference on Virtual Reality (ICVR)*, pp. 314–318, 2021. doi: 10.1109/ICVR51878.2021.9483825 1
- [3] C. Jimenez, P. Lozada, and P. Rosas. Usability heuristics: A systematic review. In *2016 IEEE 11th Colombian Computing Conference (CCC)*, pp. 1–8, 2016. doi: 10.1109/ColumbianCC.2016.7750805 1
- [4] W. Li, H. Huang, T. Solomon, B. Esmaeili, and L.-F. Yu. Synthesizing personalized construction safety training scenarios for vr training. *IEEE Transactions on Visualization and Computer Graphics*, 28(5):1993–2002, 2022. doi: 10.1109/TVCG.2022.3150510 1